

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

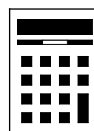
Paper 2

Wednesday 11 November 2020 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	



Answer **all** questions in the spaces provided.

Do not write
outside the
box

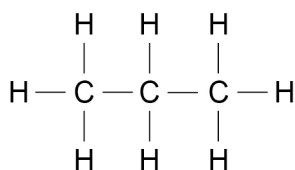
0 1

This question is about hydrocarbons.

Alkanes and alkenes are hydrocarbons.

Figure 1 represents an alkane.

Figure 1



0 1 . 1

Complete the molecular formula of the alkane shown in **Figure 1**.

[1 mark]

C ____ H ____

0 1 . 2

What is the name of the alkane in **Figure 1**?

[1 mark]

Tick (✓) **one** box.

Ethane

☐

Methane

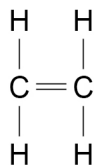
☐

Propane

☐


0 1 . 3 Figure 2 represents an alkene.

Figure 2



What is the general formula for alkenes?

[1 mark]

Tick (✓) **one** box.

C_nH_n

☐

C_nH_{2n}

☐

$\text{C}_n\text{H}_{2n-2}$

☐

$\text{C}_n\text{H}_{2n+2}$

☐

Question 1 continues on the next page

Turn over ►



A student investigated the reactions of two hydrocarbons with bromine water.

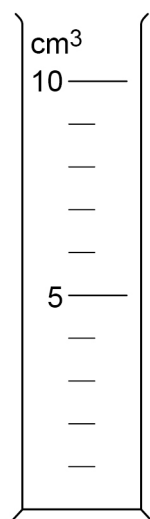
This is the method used.

- 1 Measure 2 cm³ of one of the hydrocarbons into a test tube.
- 2 Add a few drops of bromine water.
- 3 Shake the test tube.
- 4 Observe the colour of the mixture.
- 5 Repeat steps 1 to 4 using the other hydrocarbon.

0 1 . 4

Figure 3 shows one piece of equipment the student used in the experiment.

Figure 3



What is the name of this piece of equipment?

[1 mark]

Tick (✓) **one** box.

Burette

☐

Conical flask

☐

Measuring cylinder

☐

Pipette

☐

0 1 . 5

A risk assessment was done for the investigation.

The following instructions were included in the method:

- No flames in the laboratory.
- Use bromine water in a fume cupboard.

Give **one** reason for each instruction.

[2 marks]

Reason for no flames _____

Reason for using a fume cupboard _____

0 1 . 6

Which are the correct observations for the reactions between bromine water and the hydrocarbons?

Draw **one** line from each hydrocarbon to the correct observation for that hydrocarbon.

[2 marks]

Hydrocarbon	Observation
	Blue to red
	Green to colourless
Alkane	Orange to colourless
Alkene	Orange to green
	No colour change



0 2

This question is about the element titanium.

0 2 . 1An atom of titanium is represented as ${}_{22}^{48}\text{Ti}$ Give the number of protons and the number of neutrons in one atom of ${}_{22}^{48}\text{Ti}$ **[2 marks]**

Number of protons _____

Number of neutrons _____

0 2 . 2

Some titanium atoms have a different number of neutrons.

What is the name of these atoms?

[1 mark]Tick (✓) **one** box.

Ions

☐

Isotopes

☐

Molecules

☐

Polymers

☐

0 2 . 3 Titanium can be mixed with other metals.

What is the term used to describe a mixture of metals?

[1 mark]

Tick (✓) **one** box.

Alloy

☐

Compound

☐

Molecule

☐

Polymer

☐

0 2 . 4 A piece of titanium has a volume of 6.0 cm^3 and a mass of 27 g

Calculate the density of titanium in g/cm^3

Use the equation:

$$\text{Density (g/cm}^3\text{)} = \frac{\text{mass (g)}}{\text{volume (cm}^3\text{)}}$$

[2 marks]

Density = _____ g/cm^3

0 2 . 5 Titanium compounds are found in ores.

1.2% of an ore is titanium.

Calculate the mass of titanium in $5.0 \times 10^5 \text{ g}$ of the ore.

[2 marks]

Mass = _____ g

Turn over ►



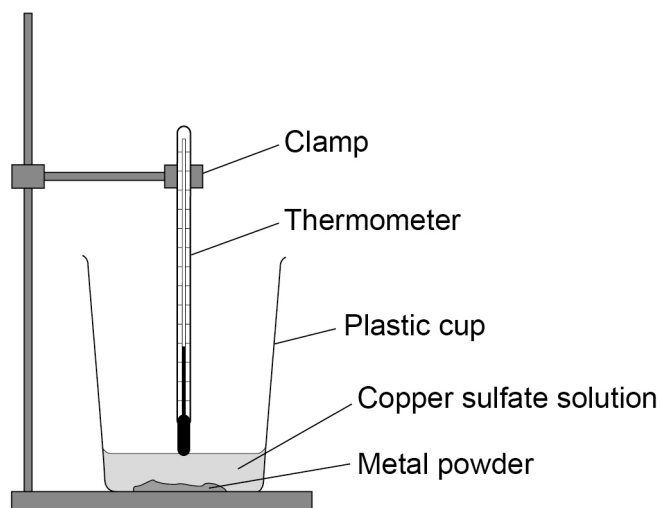
0	3
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A student investigated the reactivity of different metals using displacement reactions.

Displacement reactions are exothermic.

Figure 4 shows the apparatus used.

Figure 4



This is the method used.

- 1 Add 25 cm³ of copper sulfate solution to a plastic cup.
- 2 Measure the temperature of the copper sulfate solution.
- 3 Add 1.0 g of magnesium powder.
- 4 Record the highest temperature of the mixture.
- 5 Repeat steps 1 to 4 but use 2.0 g of zinc powder instead of 1.0 g of magnesium powder in step 3.



0 3 . 1

The apparatus and method used in the investigation did **not** give accurate results.

Suggest **three** improvements to the investigation.

[3 marks]

1 _____

2 _____

3 _____

0 3 . 2

Which of the following is a categoric variable in this investigation?

[1 mark]

Tick (✓) **one** box.

Concentration of copper sulfate solution

☐

Temperature change

☐

Type of metal

☐

Volume of copper sulfate solution

☐

Question 3 continues on the next page

Turn over ►

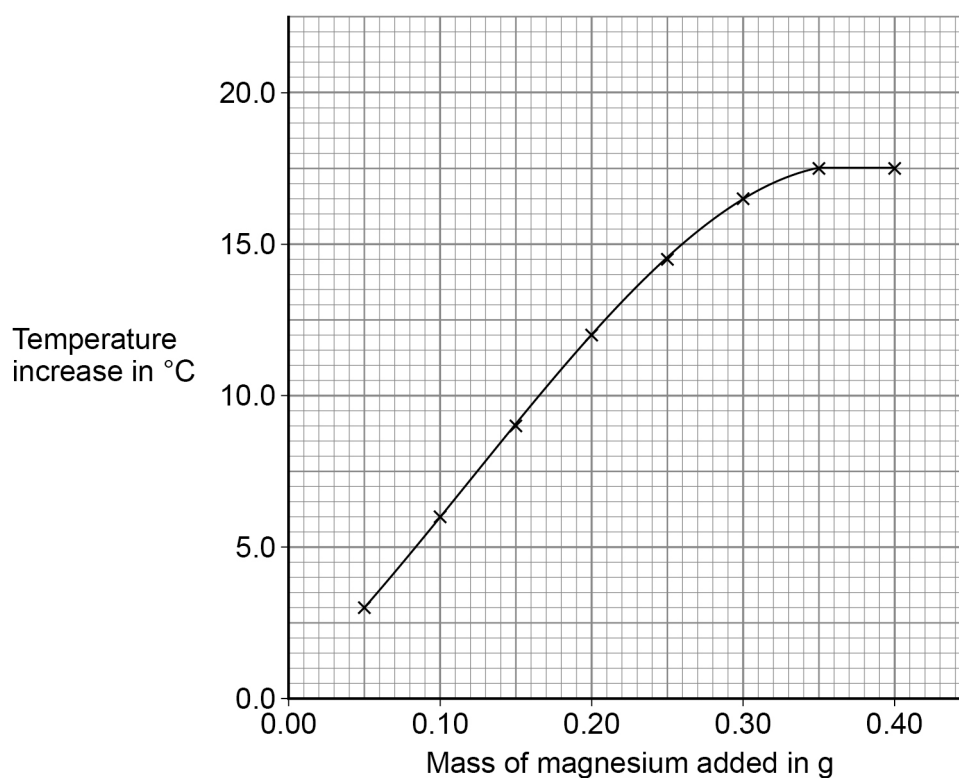
Another student did a different investigation using apparatus and a method that gave more accurate results.

This student added different masses of magnesium to 25 cm³ of copper sulfate solution.

The student measured the temperature increase.

Figure 5 shows the student's results.

Figure 5



0 3 . 3 What range of mass of magnesium did the student use in the investigation?

Use **Figure 5**.

[1 mark]

Range from _____ g to _____ g



0 3 . 4

Describe how the temperature increase changes as the mass of magnesium added increases.

Use **Figure 5**.

[3 marks]

0 3 . 5

Explain why the temperature increase in this investigation has a maximum value.

[2 marks]

10

Turn over for the next question

Turn over ►



0 4

This question is about acids and salts.

0 4 . 1

Nitric acid can be used to produce salts.

What is the formula of nitric acid?

[1 mark]Tick (✓) **one** box.CH₃COOH☐HNO₃☐H₂SO₄☐**0 4 . 2**

What is the symbol of the ion that makes solutions acidic?

[1 mark]

0 4 . 3

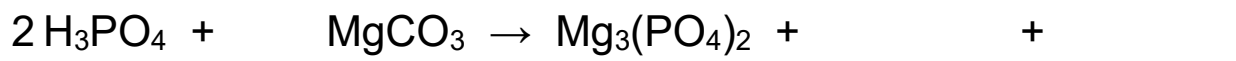
Name the salt produced when hydrochloric acid reacts with zinc oxide.

[1 mark]

0 4 . 4Phosphoric acid (H₃PO₄) reacts with magnesium carbonate to produce a salt.

Complete the equation for the reaction.

You should balance the equation.

[2 marks]

Plan a method to produce pure, dry crystals of copper sulfate from an insoluble metal oxide and a dilute acid.

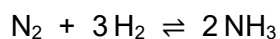
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Turn over for the next question

0 5

This question is about producing ammonia in the Haber process.

The equation for the reaction is:



This reaction is exothermic in the forward direction.

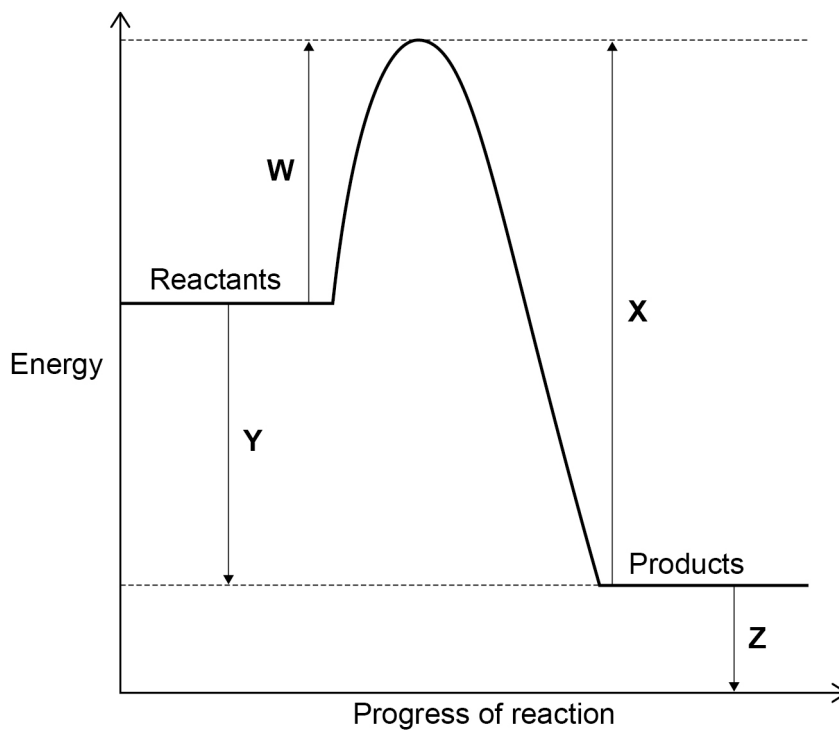
0 5**1**

Name the type of energy change in the backward direction.

[1 mark]

Figure 6 shows the energy level diagram for the reaction to produce ammonia.

Figure 6



0 5 . 2 Which arrow in **Figure 6** represents the activation energy?

[1 mark]

Tick (✓) **one** box.

W

☐

X

☐

Y

☐

Z

☐

0 5 . 3 Which arrow in **Figure 6** represents the overall energy change?

[1 mark]

Tick (✓) **one** box.

W

☐

X

☐

Y

☐

Z

☐

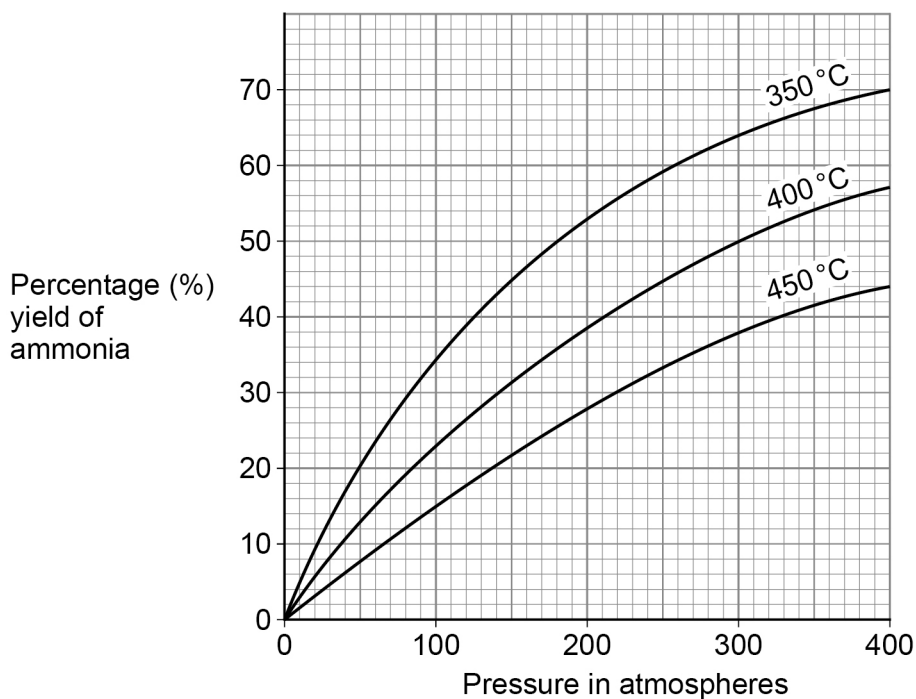
Question 5 continues on the next page

Turn over ►



Figure 7 shows the effect of pressure and temperature on the percentage yield of ammonia.

Figure 7

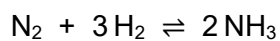


0 5 . 4

Explain the effect of pressure on the percentage yield of ammonia at 400 °C

Use **Figure 7**.

The equation for the reaction is repeated below.



[2 marks]



0 5 . 5

Which temperature and pressure would produce a 30% yield of ammonia most quickly?

Use **Figure 7**.

[2 marks]

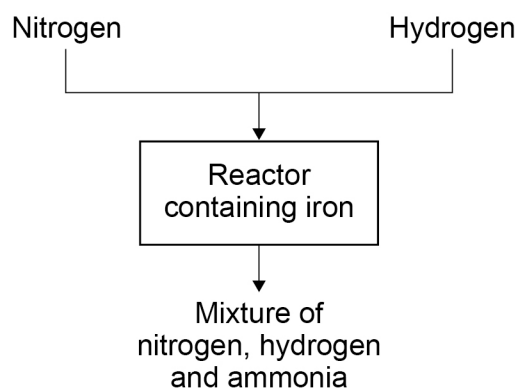
Temperature _____ °C

Pressure _____ atmospheres

0 5 . 6

Figure 8 shows an incomplete flow diagram for the production of ammonia in the Haber process.

Figure 8



Describe what happens to the mixture of nitrogen, hydrogen and ammonia after the mixture leaves the reactor.

[3 marks]

10

Turn over ►



0 6

This question is about oxides.

0 6 . 1

The formula of aluminium oxide is Al_2O_3 Calculate the relative formula mass (M_r) of aluminium oxide.Relative atomic masses (A_r): O = 16 Al = 27**[2 marks]**

Relative formula mass (M_r) of aluminium oxide = _____

0 6 . 2

Table 1 shows some properties of four different oxides, **A**, **B**, **C** and **D**.**Table 1**

Oxide	Appearance of oxide	Reaction of oxide with water
A	Colourless gas	Forms a colourless solution with pH 2
B	White solid	Forms a colourless solution with pH 1
C	White solid	Forms a colourless solution with pH 13
D	Red solid	Does not dissolve or react with water

Which oxide could be the cause of acid rain pollution?

[1 mark]Tick (✓) **one** box.
A ☐
B ☐
C ☐
D ☐

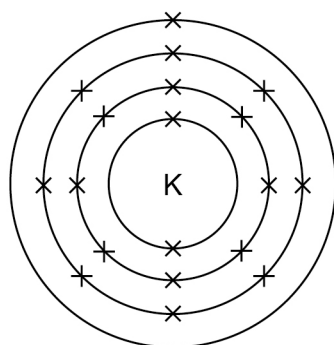

0 6 . 3

Potassium reacts with oxygen to produce an ionic compound.

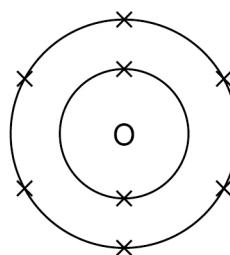
Figure 9 represents the electronic structure of a potassium atom and of an oxygen atom.

Figure 9

Potassium atom



Oxygen atom



Describe what happens when potassium reacts with oxygen to produce potassium oxide.

You should answer in terms of electrons.

[4 marks]

7

Turn over ►



07

Table 2 shows some properties of four substances, **G**, **H**, **I** and **J**.

Table 2

Substance	Melting point in °C	Electrical conductivity when solid	Electrical conductivity when liquid
G	63	Poor	Poor
H	714	Poor	Good
I	1083	Good	Good
J	3550	Poor	Poor

The four substances, **G**, **H**, **I** and **J** all have different structures.

07.1

What is the type of bonding in substance **G** and the structure of substance **G**?

Use **Table 2**.

[2 marks]

Type of bonding _____

Structure _____

07.2

Explain why substance **J** has a high melting point.

You should answer in terms of structure and bonding.

Use **Table 2**.

[3 marks]



0 7 . 3

Explain the electrical conductivity of:

- substance H
- substance I.

Use **Table 2**.**[5 marks]**

Substance H _____

Substance I _____

10**Turn over for the next question****Turn over ►**

0	8
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This question is about molecules.

0	8	.	1
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Describe the arrangement of molecules in a gas.

[2 marks]



0 8 . 2

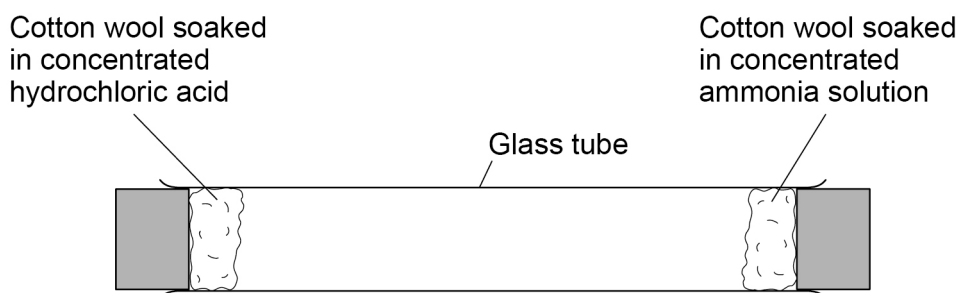
A teacher demonstrated the movement of different gas molecules.

The teacher used:

- concentrated hydrochloric acid which gives off hydrogen chloride gas
- concentrated ammonia solution which gives off ammonia gas.

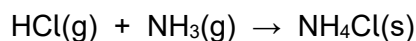
Figure 10 shows the apparatus used.

Figure 10



Hydrogen chloride gas and ammonia gas react to produce ammonium chloride, a white solid.

The equation for the reaction is:



Suggest where the white solid will form in the glass tube.

Explain your answer.

Relative formula masses (M_r): $\text{NH}_3 = 17$ $\text{HCl} = 36.5$

[4 marks]

Question 8 continues on the next page

Turn over ►



0	8	.	3
---	---	---	---

A sample of a compound contains:

- 3.6 g of carbon
- 4.2 g of nitrogen
- 1.5 g of hydrogen.

Calculate the empirical formula of the compound.

Relative atomic masses (A_r): C = 12 N = 14 H = 1

[4 marks]

Empirical formula = _____



0	8	.	4
---	---	---	---

A gas contains molecules with an empirical formula of CH_2

A sample of the gas has a volume of 4.8 dm^3 and a mass of 8.4 g

Calculate the molecular formula of the gas.

The volume of one mole of this gas is 24 dm^3

Relative atomic masses (A_r): $\text{H} = 1$ $\text{C} = 12$

[4 marks]

Molecular formula = _____

14

Turn over for the next question

Turn over ►

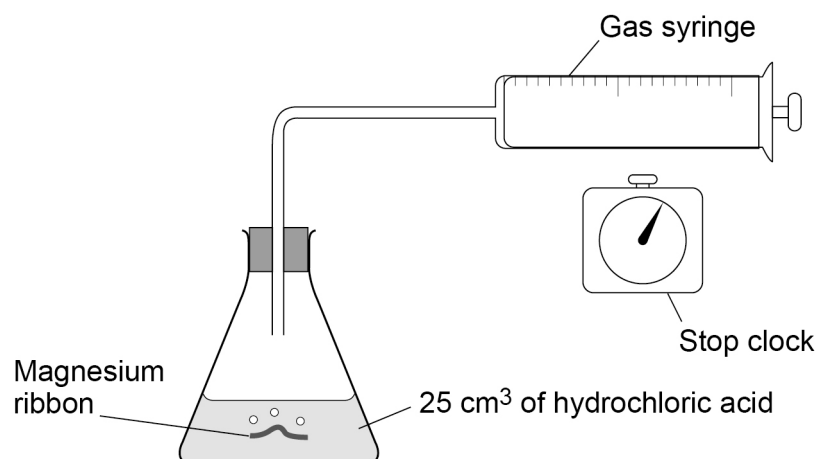


0 9

A student investigated the rate of the reaction between magnesium and hydrochloric acid.

Figure 11 shows the apparatus used.

Figure 11

**0 9****1**

Hydrogen gas is produced.

Describe the test for hydrogen gas.

Give the result if hydrogen gas is present.

[2 marks]

Test _____

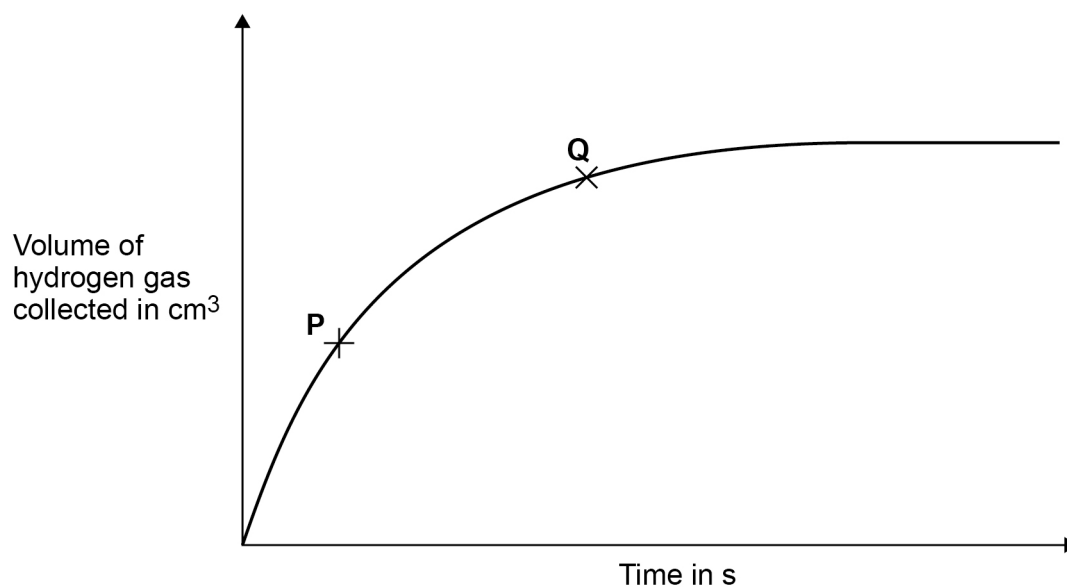
Result _____



0 9 . 2 The student measured the volume of hydrogen gas collected at regular intervals.

Figure 12 shows the results.

Figure 12



Explain what has happened to the **rate** of reaction between **P** and **Q**.

[2 marks]

Question 9 continues on the next page

Turn over ►



0 9 . 3

The student reacted 25.0 cm³ of hydrochloric acid with magnesium under different conditions.

The magnesium was in excess in these experiments.

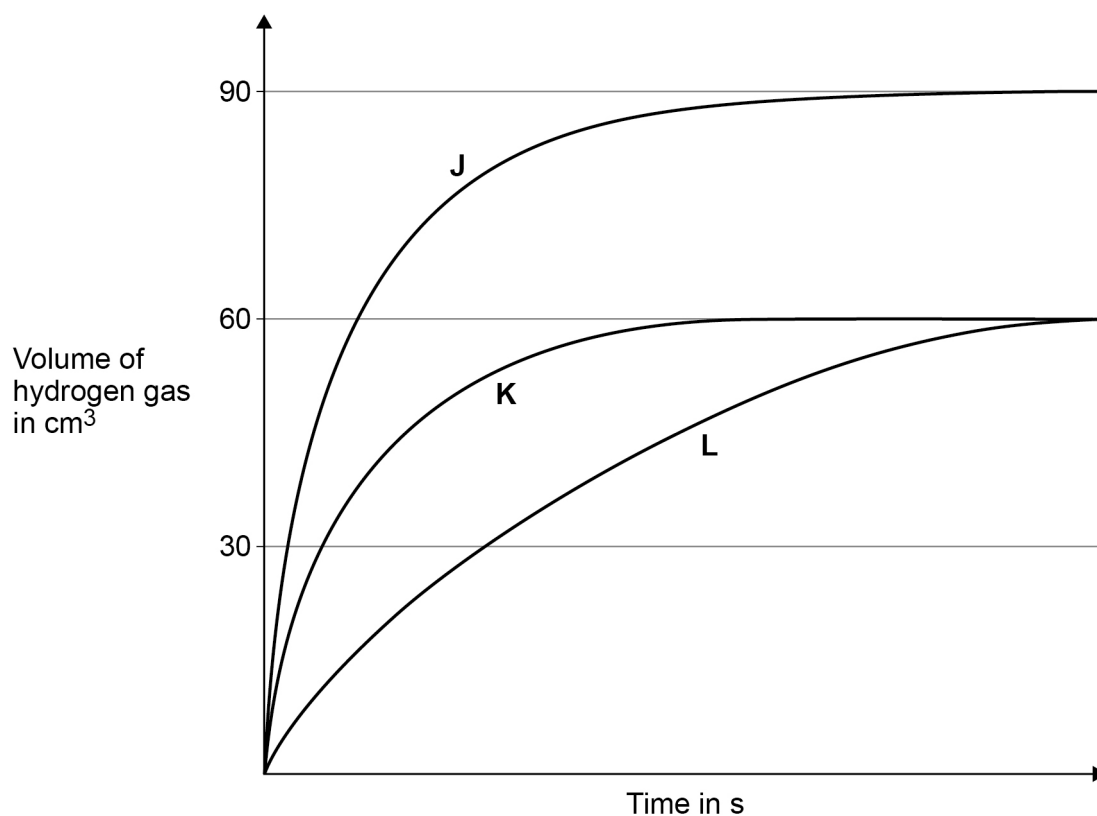
Table 3 shows the conditions used.

Table 3

Experiment	Concentration of 25.0 cm ³ hydrochloric acid in mol/dm ³	Temperature in °C
1	0.20	25
2	0.30	15
3	0.20	15

Figure 13 shows the results.

Figure 13



Which curve represents **Experiment 1**?

Tick (✓) **one** box.

Explain your choice.

[3 marks]

J ☐

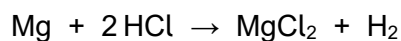
K ☐

L ☐

Explanation _____

0 9 . 4

The equation for the reaction is:



Calculate the mass of magnesium that would react completely with
25.0 cm³ of 1.5 mol/dm³ hydrochloric acid solution.

Relative atomic mass (*A_r*): Mg = 24

[5 marks]

Mass of magnesium = _____ g

12

END OF QUESTIONS



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